



Term Evaluation (Odd) Semester Examination November 2025

Roll no.....

Name of the Course: MCA AI & DS

Semester: 1st

Name of the Paper: Probability and Statistics

Paper Code: TMD 101

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions.
- (ii) Each question carries 10 marks.

Q1.

(10 Marks) CO1

a. What is data cleaning? Why is it important?

OR

b. Two coins are tossed together. Find the probability of getting

- (i) two heads
- (ii) one head
- (iii) no head

Q2.

(10 Marks) CO1

a. Define the following.

- (i) Sample space
- (ii) Independent events
- (iii) Variance
- (iv) Expectation

OR

b. Define Binomial distribution with its probability density function. If a fair coin is tossed 8 times, find the mean and variance of the number of heads obtained.

Q3.

(10 Marks) CO1 & CO2

a. A random variable has following PMF

X	0	1	2	3
P(X)	0.1	0.3	0.4	0.2

Find first four moments.

OR

b. Define mathematical expectation. Derive and prove the following two properties in case of discrete random variable

- (i) $E(X+Y) = E(X) + E(Y)$
- (ii) If X and Y are independent, then $E(XY) = E(X)E(Y)$.

Q4.

(10 Marks) CO2

a. Define normal distribution with its probability density function. For a normal distribution with mean $\mu=100$ and standard deviation 10, find the probability that a value lies between 90 and 110.
($Z=-1.00 \Rightarrow P(Z)=0.1587$, $Z=1 \Rightarrow P(Z)=0.8413$).



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OR

- b. A random variable X takes the values 1, 2, 3 with probabilities k , $2k$ and $1-3k$ respectively.

Find

- (i) the value of k ,
(ii) the mean and variance of X .

Q5.

(10 Marks) CO2

- a. Define outlier and find the outlier for the following data set of test scores of 10 students.

Scores = {75, 80, 82, 85, 78, 125, 83, 79, 81, 77}

OR

- b. Let the random variable X has the distribution:

$$P(X=0) = P(X=2) = p; \quad P(X=1) = 1-2p \quad \text{for } 0 \leq p \leq \frac{1}{2}$$

For what values of p is the $\text{Var}(X)$ a maximum.